

HOW TO FIND INTEGRATION

BY PARTIAL FRACTION



$$\int \frac{3x+5}{x^2-1} dx$$

$$\frac{2}{x-1} + \frac{1}{x+1}$$

5 STEPS



1. IDENTIFY & FACTORISE
2. SET UP PARTIAL FRACTIONS
3. FIND THE CONSTANTS
4. INTEGRATE EACH TERM
5. WRITE THE FINAL ANSWER



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YOUTUBE CHANNEL
<https://www.youtube.com/@IBMATH-in>

AHL 5.15



Content	Guidance, clarification and syllabus links
Derivatives of $\tan x$, $\sec x$, $\operatorname{cosec} x$, $\cot x$, a^x , $\log_a x$, $\arcsin x$, $\arccos x$, $\arctan x$.	
Indefinite integrals of the derivatives of any of the above functions. The composites of any of these with a linear function.	Indefinite integral interpreted as a family of curves. Examples: $\int \frac{1}{x^2 + 2x + 5} dx = \frac{1}{2} \arctan \frac{(x+1)}{2} + C$ $\int \sec^2(2x + 5) dx = \frac{1}{2} \tan(2x + 5) + C$
Use of partial fractions to rearrange the integrand.	$\int \frac{1}{x^2 + 3x + 2} dx = \ln \left \frac{x+1}{x+2} \right + C$ Link to: partial fractions (AHL1.11)



Question (1): Integrate with respect to 'x':



$$\frac{2-x}{2x^2-5x-3}$$

Let

$$\frac{2-x}{2x^2-5x-3} = \frac{2-x}{(x-3)(2x+1)} = \frac{A}{x-3} + \frac{B}{2x+1}$$

$$\therefore 2-x = A(2x+1) + B(x-3) \text{ —————(i)}$$

Put $x = 3$ in (i), We get

$$2 - (3) = A(2(3) + 1) + B(0)$$

$$\therefore A = -\frac{1}{7}$$

Put $x = -1/2$ in (i), We get

$$2 - \left(-\frac{1}{2}\right) = A(0) + B\left(-\frac{1}{2} - 3\right) \Rightarrow B = -\frac{5}{7}$$

$$\therefore \frac{2-x}{2x^2-5x-3} = \frac{-1/7}{x-3} + \frac{-5/7}{2x+1}$$

$$\left\{ \begin{array}{l} 2x^2 - 5x - 3 \\ = 2x^2 - 6x + x - 3 \\ = 2x(x-3) + 1(x-3) \\ = (x-3)(2x+1) \end{array} \right\}$$



$$\therefore \frac{2-x}{2x^2-5x-3} = \frac{-1/7}{x-3} + \frac{-5/7}{2x+1}$$

$$\therefore \int \frac{2-x}{2x^2-5x-3} dx = \int \frac{-1/7}{x-3} dx + \int \frac{-5/7}{2x+1} dx$$

$$= -1/7 \ln |x-3| - 5/7 \frac{\ln |2x+1|}{2} + C$$

$$= -\frac{1}{7} \ln |x-3| - \frac{5}{14} \ln |2x+1| + C$$

Question (2): Integrate with respect to 'x':



a Write $\frac{6x^2 + x - 19}{(x+3)(x-1)^2}$ as the sum of partial fractions in the form $\frac{A}{x+3} + \frac{B}{x-1} + \frac{C}{(x-1)^2}$.

b Hence find $\int \frac{6x^2 + x - 19}{(x+3)(x-1)^2} dx$.

(a) Let
$$\frac{6x^2 + x - 19}{(x+3)(x-1)^2} = \frac{A}{x+3} + \frac{B}{x-1} + \frac{C}{(x-1)^2}.$$

$$\therefore 6x^2 + x - 19 = A(x-1)^2 + B(x+3)(x-1) + C(x+3) \text{ --- (i)}$$

Put $x = -3$ in (i), We get

$$6(-3)^2 + (-3) - 19 = A(-3-1)^2 + B(0) + C(0)$$

$$32 = 16A \Rightarrow A = 2$$

Put $x = 1$ in (i), We get

$$6(1)^2 + (1) - 19 = A(0) + B(0) + C(1+3)$$

$$\therefore C = -3$$



$$\therefore 6x^2 + x - 19 = A(x-1)^2 + B(x+3)(x-1) + C(x+3) \text{ --- (i)}$$

Put $x = 0$ in (i), We get

$$6(0)^2 + (0) - 19 = A(0-1)^2 + B(0+3)(0-1) + C(0+3)$$

$$-19 = (2) - 3B + (-3)(3)$$

$$\therefore B = 4$$

$$\therefore \frac{6x^2 + x - 19}{(x+3)(x-1)^2} = \frac{2}{x+3} + \frac{4}{x-1} - \frac{3}{(x-1)^2}$$

$$\begin{aligned} \text{(b)} \quad \therefore \int \frac{6x^2 + x - 19}{(x+3)(x-1)^2} dx &= \int \frac{2}{x+3} dx + \int \frac{4}{x-1} dx - \int \frac{3}{(x-1)^2} dx \\ &= \int \frac{2}{x+3} dx + \int \frac{4}{x-1} dx - \int 3(x-1)^{-2} dx \\ &= 2 \ln |x+3| + 4 \ln |x-1| - 3 \frac{(x-1)^{-1}}{-1} + C \\ &= 2 \ln |x+3| + 4 \ln |x-1| + \frac{3}{x-1} + C \end{aligned}$$

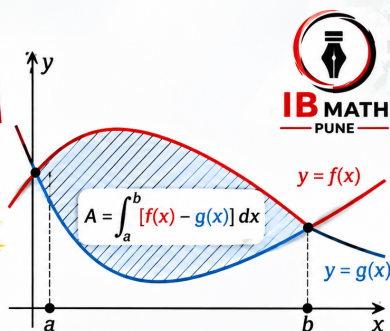


Practice questions PDF link is given in the description of this video.



HOW TO FIND THE AREA BETWEEN TWO CURVES IN 5 EASY STEPS

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